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Methodology

Valuation of Multidimensional Health States With a Bolt-On: Is There a Shortcut?

Zhihao Yang, PhD, Kim Rand, PhD, Nan Luo, PhD

ABSTRACT

Objectives: To use the EQ-5D questionnaire with bolt-on dimensions in economic evaluation studies, new value sets are needed. In this study, we explored the feasibility of a new approach called the scaling factor model, which estimates bolt-on value sets using estimated EQ-5D dimensional weights.

Methods: We designed a 2-arm study, inviting university students to value health states with and without bolt-on items using the composite time trade-off method. We selected 25 health states from an orthogonal array and added the 5 mildest EQ-5D states in the design. In arm 1, EQ-5D without self-care and standard EQ-5D states were valued, and in arm 2, standard EQ-5D states and EQ-5D with vision were valued. By arm, we compared the mean observed values of health states with and without bolt-on item. Next, by arm, we estimated value sets for the EQ-5D with bolt-on states using both standard model and scaling factor model. Model performances were compared in terms of prediction accuracy and correlation with likelihood-based mean values.

Results: Adding a five-level bolt-on to EQ-5D resulted in statistically lower values. This effect was consistent across 2 arms and bolt-on items. The scaling factor models outperformed the standard models in all statistics.

Conclusions: The scaling factor model offers a methodologically viable and low-cost option for producing value sets for EQ-5D supplemented with bolt-on items. Future studies should further test this method using other bolt-on items and more relevant study populations.

Keywords: bolt-on, EQ-5D-5L, scaling factor model, value set.

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Highlights

- Values for the EQ-5D with a bolt-on item cannot be derived from standard EQ-5D value sets. In contrast to the traditional method of independently estimating the value set, this study introduces an innovative approach called scaling factor model to estimate value sets for EQ-5D health states with a bolt-on item.
- Compared with the standard modeling approach, the scaling factor model predicts the values of bolt-on health states more accurately. This finding suggests that bolt-on value sets can be estimated based on the standard EQ-5D value sets, thereby enhancing resource efficiency and maintaining dominance relationships among health states.
- The use of the scaling factor model provides a cost-effective means to estimating the value set for EQ-5D health states with a bolt-on item. Another advantage of the scaling factor model is that it avoids potential inconsistencies between the value sets of original EQ-5D states and EQ-5D states with bolt-on items.

Introduction

The EQ-5D is an instrument used to measure health-related quality of life (HRQoL) in 5 dimensions: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Although it is widely used to compare HRQoL across different disease and intervention groups, studies have shown that it may not be very sensitive to certain health conditions, such as hearing, vision, and cognition impairments.^{1–3} To address this issue, researchers have proposed adding new dimensions to the existing EQ-5D system, also known as bolt-on items, to improve its content validity while preserving the core components. For instance, it has been found that vision and hearing bolt-on items improved the sensitivity of EQ-5D in assessing the HRQoL of individuals with vision and hearing problems, respectively.^{1,2}

Expanding the descriptive system of EQ-5D with new dimensions renders the values of derived health states inaccessible through existing EQ-5D value sets. Thus, new value sets that incorporate the effect of the bolt-on items are required. Such new value sets can be developed using the EQ-VT protocol.⁴ However, new, full-scale value set studies are expensive undertakings that involve computer-assisted personal interviewing of large general population samples.⁵ In particular, because the

current practice is for each country to develop and use EQ-5D value sets estimated using health preferences of its local public, country-specific bolt-on value sets are likely to be required. This approach would be very burdensome considering the large and ever-increasing number of bolt-on items. The high costs of developing bolt-on value sets in this way might prevent the use of bolt-on items in economic evaluations where EQ-5D is deemed an inadequate HRQoL measure.

Moreover, values estimated from new, stand-alone bolt-on valuation studies may be inconsistent with existing EQ-5D values because of the possible interaction between the bolt-on item and the 5 core dimensions, sampling method, and/or interviewer effects.⁶ For example, it is possible that EQ-5D health-state A has a higher utility value than B according to an EQ-5D value set, but A has a lower value than B after the 2 health states are extended with a vision bolt-on (eg, slight problems in seeing things). Such inconsistencies between standard and bolt-on EQ-5D values, if

existent, may reverse economic evaluation results when investigators switch the value set for calculating quality-adjusted life-years. Thus, an approach that can ensure the consistency between existing EQ-5D and new bolt-on value sets would be preferred, although inconsistency due to true preference change is theoretically possible.

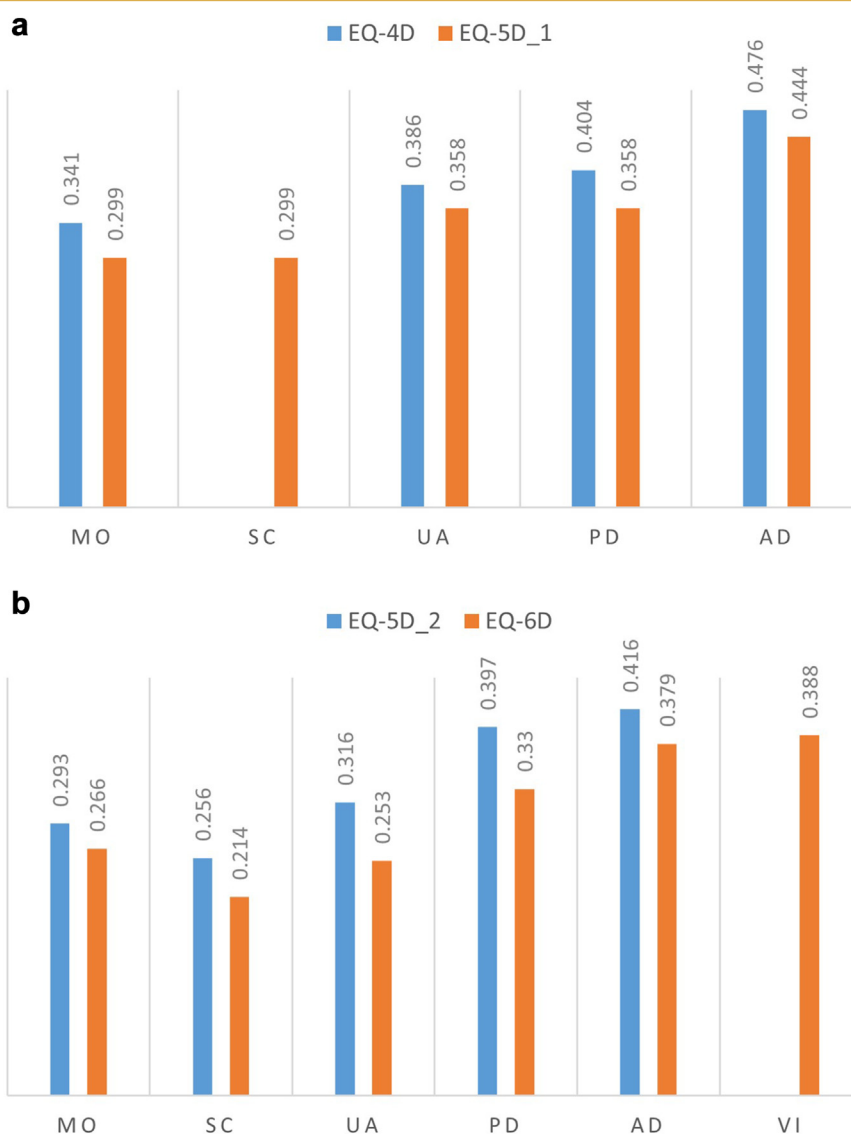
The above-mentioned concerns could be addressed if bolt-on value sets could be linked to existing EQ-5D value sets in some way to leverage the relative importance of the 5 core dimensions from standard EQ-5D value sets and only estimate the effect of bolt-on items. This approach seems feasible because the relative importance of the 5 EQ-5D dimensions estimated before and after adding the vision bolt-on was similar, but all decreased proportionally in our recent work (see Fig. 1, second arm). A similar result was observed when treating self-care as a bolt-on to the remaining 4 EQ-5D dimensions (Fig. 1, first arm).⁷ This observation suggests that the relative importance of the 5 core dimensions may be preserved after adding the bolt-on item, but the absolute

values of these parameters decrease. Therefore, it seems that we may use the disutility weights of existing EQ-5D value sets to estimate bolt-on values, after accounting for the level of decrease in all dimensions.

A major advantage of this linking approach is that only a small valuation study is required, focusing on the estimation of the disutility of the bolt-on items relative to the disutility of the 5 core dimensions. The disutility of the 5 core dimensions could be taken from the published, standard EQ-5D value set. This is a potentially cost-saving approach compared with the full-scale valuation study design. This method also ensures consistency between bolt-on value sets and standard EQ-5D value sets, which avoids the possibility of gaming.

In this study, our aim was to investigate the feasibility of this linking approach by reanalyzing data collected in a previous valuation study in which composite time trade-off (cTTO) data were collected for both standard and altered EQ-5D health states.

Figure 1. Bar plots for coefficients of models with and without bolt-on dimension. (A) First arm. (B) Second arm.



AD indicates anxiety/depression; MO, mobility; PD, pain/discomfort; SC, self-care; UA, usual activities; VI, vision.

Methods

Experimental Design

We conducted an experiment to value EQ-5D-5L health states (hereafter referred to as EQ-5D), vision bolt-on health states (EQ-5D-5L health states with a vision bolt-on, hereafter referred to as EQ-6D), and modified EQ-5D-5L health states (EQ-5D-5L health states without the self-care dimension, hereafter referred to as EQ-4D). This design was intended to investigate whether the effect of adding a dimension to a descriptive system varies with the number of dimensions in the descriptive system. Given the exploratory nature of this study, we selected vision and self-care as bolt-on items because previous research has indicated that vision can improve the discrimination power of EQ-5D,¹ whereas self-care is the EQ-5D dimension with the least reported problems for the Chinese population.⁸ We collected cTTO values from 600 university students, in line with sample size considerations proposed by Oppe et al.⁴

The valuation study was conducted at Guizhou Medical University in China with a sample of students. The students were recruited through advertisements and word of mouth and were randomly assigned to 2 arms for face-to-face interviews using the cTTO protocol implemented by EQ-PVT, a Microsoft PowerPoint-based version of EQ-VT v1.1.⁵

The Interviews

The data collection process was described in detail elsewhere.⁷ In brief, a team of 8 interviewers who had experience conducting cTTO interviews were recruited and re-trained. EQ-PVT was used,⁹ and the standard quality control procedure of the EuroQol Group¹⁰ was followed. To ensure protocol compliance and identify any interviewer effects, quality control reports were generated every 10 interviews and sent to the interviewers. If necessary, individual suggestions were provided to improve interview performance. Participants were recruited in several schools of Guizhou Medical University using social media, and an appointment was set up by a study coordinator for the interview.

After giving consent and self-reporting health using EQ-5D (EQ-4D for respondents in arm 1, EQ-5D for respondents in arm 2) and EQ-VAS, respondents valued a randomly selected block of 20 health states, which included 10 EQ-5D health states and 10 EQ-4D/EQ-6D health states. Respondents in the first arm were given 3 wheelchair examples (being in a wheelchair, a state better than being in a wheelchair, a state worse than being in a wheelchair), followed by 3 EQ-4D health states for practice, 10 EQ-4D health states, and 10 EQ-5D health states, in that order. Similarly, respondents in the second arm were given 3 wheelchair examples, followed by 3 EQ-5D health states for practice, 10 EQ-5D health states, and 10 EQ-6D health states for valuation, in that order. The 3 wheelchair examples were used to familiarize respondents with the cTTO procedure, including the part for valuing health states that respondents may consider worse than death. The 3 practice states were selected to represent different severity levels (2121, 3554, and 1411 for the first arm and 21121, 35554, and 15411 for the second arm). No feedback or debriefing module was provided to respondents. Respondents always valued health states with fewer dimensions first (ie, EQ-4D for arm 1 and EQ-5D for arm 2) and did not see the bolt-on health states until they had finished the valuation of the health states with fewer dimensions.

Health-States Selection

Previous research has demonstrated that an orthogonal design comprising 25 health states can be utilized to estimate an EQ-5D

health-state value set.¹¹ To estimate the EQ-4D, EQ-5D, and EQ-6D value sets, we adopted an existing orthogonal array of 6×5 .¹² This orthogonal array contained 6 columns, each representing a dimension, and could therefore be utilized to represent EQ-6D states. To derive the EQ-4D states, we removed the second and last columns of the orthogonal array. Similarly, to derive the EQ-5D states, we removed the last column of the orthogonal array. One notable characteristic of an orthogonal design is that the remaining columns remain orthogonal when one or more columns are removed. From all the variants of designs that we created, we selected the one that contained the state 55555 and the most plausible states.¹³ Furthermore, we added all 5 mildest states in the design.⁹ Using this approach, our design consisted of 29 EQ-4D states, 30 EQ-5D states, and 31 EQ-6D states. Because all the states were derived from the same orthogonal array, the levels of core dimensions (ie, mobility, usual activities, pain/depression, and anxiety/depression) across the 3 descriptive systems were the same.

We divided all 90 states into 6 blocks, each comprising 2 types of states. First, using the Alg-algorithm in R, we divided the EQ-6D, EQ-4D, and EQ-5D states into 3 blocks each. Next, we combined the 3 EQ-4D blocks with their corresponding EQ-5D blocks to create block 1, 2, and 3, each containing 10 EQ-4D states and 10 EQ-5D states for valuation in the first arm. Similarly, we created block 4, 5, and 6, each containing 10 EQ-5D states and 10 EQ-6D states, for valuation in the second arm. A strength of the two-arm design lies in better understanding the effect of adding different bolt-on items (self-care and vision), which could vary in both types of health states (EQ-4D and EQ-5D) with a different number of core dimensions (4 and 5).

Data Analysis

Data collected from the 2 arms were analyzed separately using the same method. To examine how adding a bolt-on item could affect the cTTO values, we calculated the mean cTTO values by health state with and without the bolt-on item and compared them using a paired *t*-test. Next, we modeled the cTTO data by type of health states. For EQ-4D and EQ-5D (arm 1), as well as for EQ-5D (arm 2) and EQ-6D, we estimated and compared the performance of the conventional main-effects model and the scaling factor model, which included the estimated model coefficients for EQ-4D or EQ-5D (arm 2). Below we explained the modeling process in detail.

We estimated 4 conventional models, 1 for each type of health state from each arm, using both the additive model and the cross-attribute level effects (CALE) model specifications. In a CALE model, a single parameter is estimated per dimension, conventionally taking the value of the worst level, and cross-attribute coefficients for the levels (eg, level 2 “slight,” level 3 “moderate,” and level 4 “severe”) are used. This format produces a constrained variation compared with the regular models, in which a single parameter is estimated per level and dimension combination. CALE models have been found to be superior to corresponding additive models in terms of out-of-sample predictive accuracy when modeling EQ-5D^{14,15} and modified EQ-5D health states.⁷ By leveraging the similarity in level labels across EQ-5D dimensions, CALE models decrease the number of model parameters. For example, an 8-parameter model can be constructed for EQ-5D, including 5-dimension parameters, each representing the level-5 problems of a different dimension, and 3 level parameters for slight (level-2), moderate (level-3), and severe (level-4). The model operates under the assumption that the ratios of disutility associated with level-2 to level-4 problems versus level-5 problems are constant across dimensions. These models are also referred to as multiplicative models because the disutility for each dimension at each level is calculated by multiplying a level parameter by a dimension parameter.

These 4 models include a 7-parameter model using EQ-4D data from arm 1 (model 4D, formula 1), an 8-parameter model using EQ-5D data from arm 1 (model 5D_ARM1, formula 2), an 8-parameter model using EQ-5D data from arm 2 (model 5D_ARM2, formula 2), and a 9-parameter model using EQ-6D data from arm 2 (model 6D, formula 3).

Supplemental Materials found at <https://doi.org/10.1016/j.jval.2024.06.018> includes the overview of all models.

Next, we used a by-state cross-validation approach^{7,14} to evaluate the performance of the scaling factor models and the conventional models in predicting observed, out-of-sample health-state values. In brief, we left 1 health state out and

$$\begin{aligned} \text{EQ-4D disutility} = & \alpha + (\beta_{MO}x_{MO2} + \beta_{UA}x_{UA2} + \beta_{PD}x_{PD2} + \beta_{AD}x_{AD2})\beta_{L2} + (\beta_{MO}x_{MO3} + \beta_{UA}x_{UA3} + \beta_{PD}x_{PD3} + \beta_{AD}x_{AD3})\beta_{L3} + \\ & (\beta_{MO}x_{MO4} + \beta_{UA}x_{UA4} + \beta_{PD}x_{PD4} + \beta_{AD}x_{AD4})\beta_{L4} + (\beta_{MO}x_{MO5} + \beta_{UA}x_{UA5} + \beta_{PD}x_{PD5} + \beta_{AD}x_{AD5}) + \epsilon \end{aligned} \quad (1)$$

$$\begin{aligned} \text{EQ-5D disutility} = & \alpha + (\beta_{MO}x_{MO2} + \beta_{SC}x_{SC2} + \beta_{UA}x_{UA2} + \beta_{PD}x_{PD2} + \beta_{AD}x_{AD2})\beta_{L2} + (\beta_{MO}x_{MO3} + \beta_{SC}x_{SC3} + \beta_{UA}x_{UA3} + \beta_{PD}x_{PD3} + \\ & \beta_{AD}x_{AD3})\beta_{L3} + (\beta_{MO}x_{MO4} + \beta_{SC}x_{SC4} + \beta_{UA}x_{UA4} + \beta_{PD}x_{PD4} + \beta_{AD}x_{AD4})\beta_{L4} + (\beta_{MO}x_{MO5} + \beta_{SC}x_{SC5} + \beta_{UA}x_{UA5} + \beta_{PD}x_{PD5} + \beta_{AD}x_{AD5}) + \epsilon \end{aligned} \quad (2)$$

$$\begin{aligned} \text{EQ-6D disutility} = & \alpha + (\beta_{MO}x_{MO2} + \beta_{SC}x_{SC2} + \beta_{UA}x_{UA2} + \beta_{PD}x_{PD2} + \beta_{AD}x_{AD2} + \beta_{VI}x_{VI2})\beta_{L2} + (\beta_{MO}x_{MO3} + \beta_{SC}x_{SC3} + \beta_{UA}x_{UA3} + \beta_{PD}x_{PD3} + \beta_{AD}x_{AD3} + \\ & \beta_{VI}x_{VI3})\beta_{L3} + (\beta_{MO}x_{MO4} + \beta_{SC}x_{SC4} + \beta_{UA}x_{UA4} + \beta_{PD}x_{PD4} + \beta_{AD}x_{AD4} + \beta_{VI}x_{VI4})\beta_{L4} + (\beta_{MO}x_{MO5} + \beta_{SC}x_{SC5} + \beta_{UA}x_{UA5} + \beta_{PD}x_{PD5} + \beta_{AD}x_{AD5} + \beta_{VI}x_{VI5}) + \epsilon \end{aligned} \quad (3)$$

We constructed a scaling factor model based on the observation that, after adding a bolt-on item, the value of all dimensional coefficients estimated using the above-described models decreased to a similar degree (between 6.72% and 12.3% for the first arm, between 8.89% and 19.9% for the second arm, see Fig. 1). It seemed that dimensional disutility weights shrink with the same proportion when a bolt-on item is added to a multi-dimensional health state. Therefore, we constructed the scaling factor model for predicting the values of bolt-on health states by converting relevant dimensional coefficients of the additive or CALE model to constants using estimated dimensional disutility and adding a scaling factor into the model to account for the shrinkage of the constants. For example, in arm 1, we used the coefficients estimated in the 7-parameter model (model 4D) to replace β_{MO} , β_{UA} , β_{PD} , β_{AD} , β_{L2} , β_{L3} , and β_{L4} in the model for predicting EQ-5D values. This scaling factor model only estimates β_{SC} , α (INTERCEPT), and λ (scaling factor) (formula 4). Similarly, for arm 2, the scaling factor model for EQ-6D uses coefficients estimated in model 5D_ARM2 to replace β_{MO} , β_{SC} , β_{UA} , β_{PD} , β_{AD} , β_{L2} , β_{L3} , and β_{L4} , and only estimates β_{VI} , α , and λ (formula 5). We estimated the scaling factor model using both additive and CALE model specifications. All models were specified to have a random-effects intercept at the level of individual study participants, using incremental dummies and to be censored at -1 . Appendix 1 in

estimated the models using the cTTO values of the remaining health states; we then used the model coefficients to estimate the value of the left-out state. We repeated this analysis for all the health states. For each health state, we used the likelihood-based mean value predicted from a Tobit model as the observed cTTO value because the observed value was censored at -1 . Using the observed values and predicted values, we computed Pearson product-moment correlation (R), Lin concordance correlation coefficient, mean absolute errors (MAE), and root mean squared error and used them to compare the performance of the conventional and scaling factor models.

Results

A total of 600 students participated in the study and completed the valuation tasks. On average, each student spent 36.7 minutes (SD: 10.8 minutes) completing all the cTTO tasks, including the 6 practice tasks. On average, each student took 6.77 moves (SD: 1.97 moves) to reach the indifferent point. The histograms of the cTTO values by arm and type of health states were reported elsewhere.⁷ Overall, the values were distributed similarly across arms and types of health states. In all histograms, there was a spike at -1 ,

$$\begin{aligned} \text{EQ-5D disutility} = & \alpha + \lambda ([\beta_{MO}x_{MO2} + \beta_{SC}x_{SC2} + \beta_{UA}x_{UA2} + \beta_{PD}x_{PD2} + \beta_{AD}x_{AD2}]\beta_{L2} + [\beta_{MO}x_{MO3} + \beta_{SC}x_{SC3} + \beta_{UA}x_{UA3} + \beta_{PD}x_{PD3} + \beta_{AD}x_{AD3}]\beta_{L3} + \\ & [\beta_{MO}x_{MO4} + \beta_{SC}x_{SC4} + \beta_{UA}x_{UA4} + \beta_{PD}x_{PD4} + \beta_{AD}x_{AD4}]\beta_{L4} + [\beta_{MO}x_{MO5} + \beta_{SC}x_{SC5} + \beta_{UA}x_{UA5} + \beta_{PD}x_{PD5} + \beta_{AD}x_{AD5}]) + \epsilon \end{aligned} \quad (4)$$

$$\begin{aligned} \text{EQ-6D disutility} = & \alpha + \lambda ([\beta_{MO}x_{MO2} + \beta_{SC}x_{SC2} + \beta_{UA}x_{UA2} + \beta_{PD}x_{PD2} + \beta_{AD}x_{AD2} + \beta_{VI}x_{VI2}]\beta_{L2} + [\beta_{MO}x_{MO3} + \beta_{SC}x_{SC3} + \beta_{UA}x_{UA3} + \beta_{PD}x_{PD3} + \beta_{AD}x_{AD3} + \\ & \beta_{VI}x_{VI3}]\beta_{L3} + [\beta_{MO}x_{MO4} + \beta_{SC}x_{SC4} + \beta_{UA}x_{UA4} + \beta_{PD}x_{PD4} + \beta_{AD}x_{AD4} + \beta_{VI}x_{VI4}]\beta_{L4} + [\beta_{MO}x_{MO5} + \beta_{SC}x_{SC5} + \beta_{UA}x_{UA5} + \beta_{PD}x_{PD5} + \beta_{AD}x_{AD5} + \beta_{VI}x_{VI5}]) + \epsilon \end{aligned} \quad (5)$$

Table 1. EQ-4D and EQ-5D health-state values from arm 1.

EQ-4D state	Mean	SD	EQ-5D state	Mean	SD	Self-care level	Mean changes
1*112	0.928	0.095	11112	0.946	0.072	1	0.018
1*121	0.936	0.061	11121	0.943	0.053	1	0.007
1*211	0.899	0.104	11211	0.910	0.092	1	0.012
2*111	0.901	0.131	21111	0.918	0.109	1	0.017
1*453	0.146	0.474	11453	0.298	0.428	1	0.152 [†]
2*514	0.171	0.542	21514	0.320	0.483	1	0.149 [†]
3*125	0.313	0.481	31125	0.428	0.413	1	0.114
4*231	0.546	0.274	41231	0.573	0.322	1	0.027
5*342	0.168	0.502	51342	0.250	0.494	1	0.082
2*245	0.091	0.521	22245	0.147	0.548	2	0.056
3*533	0.234	0.469	32533	0.342	0.417	2	0.108
4*354	−0.134	0.518	42354	0.045	0.524	2	0.180 [†]
5*421	0.278	0.452	52421	0.397	0.419	2	0.119
/	/	/	12111	0.912	0.124	2	/
/	/	/	12112	0.877	0.105	2	/
1*541	0.294	0.445	13541	0.347	0.418	3	0.053
2*323	0.612	0.342	23323	0.619	0.216	3	0.007
3*252	0.371	0.449	33252	0.363	0.427	3	−0.008
4*415	−0.071	0.561	43415	−0.001	0.555	3	0.071
5*134	0.158	0.513	53134	0.218	0.476	3	0.059
1*335	0.317	0.465	14335	0.129	0.518	4	−0.188 [†]
2*151	0.462	0.433	24151	0.379	0.396	4	−0.083
3*444	−0.098	0.542	34444	−0.257	0.519	4	−0.158 [†]
4*522	0.211	0.535	44522	−0.013	0.579	4	−0.224 [†]
5*213	0.468	0.350	54213	0.365	0.403	4	−0.103
1*224	0.494	0.382	15224	0.319	0.461	5	−0.175 [†]
2*432	0.458	0.397	25432	0.312	0.441	5	−0.147 [†]
3*311	0.713	0.164	35311	0.547	0.258	5	−0.166 [†]
4*143	0.250	0.453	45143	0.130	0.478	5	−0.120
5*555	−0.592	0.414	55555	−0.682	0.379	5	−0.090 [†]

*For EQ-4D states, self-care dimension (the second digital of standard EQ-5D health state) was removed.

[†]Significant at 0.05 level.

which suggests that some values may be lower than −1 if the cTTO procedure had allowed for such values.

In arm 1 (see Table 1), the mean observed value for all EQ-4D states was 0.278 (SE: 0.011) and the mean observed value for all EQ-5D states was 0.264 (SE: 0.011). EQ-4D had higher values and the difference was not significant at the 0.05 level. In arm 2 (see Table 2), the mean observed value for all EQ-5D states was 0.305 (SE: 0.010), and the mean observed value for all EQ-6D states was 0.270 (SE: 0.011). The difference was significant. Tables 1 and 2 display the mean values of the health states and the level of bolt-on for 2 arms, respectively. Overall, the tables demonstrate that adding a bolt-on dimension to EQ-5D resulted in increased values when the levels of the bolt-on dimension were between 1 (no problems) and 3 (moderate problems). Notably, the increment was only significant at the 0.05 level for 3 health-state pairs (EQ-4D health state: 1453, 2514, and 4354) for the first arm. All increments did not reach statistical significance in the second arm. In comparison, values decreased when the levels were at 4 (severe problems) or 5 (extreme problems), and most decrements were

statistically significant. The effect was different in these 2 arms. Adding severe problems (level 4) self-care bolt-on to the health state resulted large decrease in observed values, but this was not the case for the severe problems (level 4) of the vision bolt-on. All decrements for severe problems of the vision bolt-on were not significant. Adding a level 5 bolt-on always resulted in a significant decrease in observed value for both arms, except for the health-state 4143 from the first arm. Overall, the increment and decrement were dependent on the severity of the core dimensions, with a smaller increment for better core dimensions and a smaller decrement for worse core dimensions. These findings suggest that bolt-on dimensions had an effect on the overall health-state values, expanding the range of values toward both ends of the scale, with the effect more obvious toward the lower end of the scale.

Table 3 displays the estimated coefficients and model performance results for the additive models. Almost all coefficients estimated for most of the main effects show monotonicity. Non-monotonicity was only observed for the coefficient when

Table 2. EQ-5D and EQ-6D health-state values from arm 2.

EQ-5D state	Mean	SD	EQ-6D state	Mean	SD	Vision level	Mean changes
11112*	0.939	0.075	111121	0.952	0.062	1	0.013
11121*	0.929	0.100	111211	0.946	0.062	1	0.018
11211*	0.905	0.105	112111	0.927	0.080	1	0.022
12111*	0.923	0.092	121111	0.934	0.074	1	0.011
21111*	0.919	0.098	211111	0.936	0.063	1	0.017
15224*	0.377	0.436	152241	0.414	0.450	1	0.036
24151*	0.421	0.361	241511	0.437	0.349	1	0.016
32533*	0.310	0.441	325331	0.417	0.349	1	0.107
43415*	0.115	0.469	434151	0.180	0.446	1	0.066
51342*	0.326	0.381	513421	0.338	0.399	1	0.012
11453*	0.172	0.547	114532	0.249	0.516	2	0.077
22245*	0.110	0.578	222452	0.162	0.581	2	0.052
35311*	0.497	0.356	353112	0.562	0.346	2	0.066
44522*	0.148	0.476	445222	0.218	0.431	2	0.070
53134*	0.152	0.565	531342	0.144	0.538	2	−0.008
/	/	/	111112	0.969	0.044	2	/
14335*	0.209	0.426	143353	0.262	0.394	3	0.053
21514*	0.288	0.492	215143	0.319	0.462	3	0.031
33252*	0.384	0.398	332523	0.471	0.298	3	0.087
45143*	−0.005	0.552	451433	0.092	0.537	3	0.097
52421*	0.407	0.446	524213	0.432	0.350	3	0.025
13541*	0.236	0.525	135414	0.179	0.521	4	−0.057
25432*	0.369	0.448	254324	0.267	0.445	4	−0.102
31125*	0.415	0.440	311254	0.326	0.457	4	−0.089
42354*	−0.026	0.544	423544	−0.052	0.547	4	−0.026
54213*	0.285	0.479	542134	0.178	0.513	4	−0.107
12112*	0.887	0.101	121125	0.450	0.394	5	−0.437 [†]
23323*	0.624	0.229	233235	0.237	0.497	5	−0.387 [†]
34444*	−0.165	0.462	344445	−0.415	0.450	5	−0.250 [†]
41231*	0.609	0.208	412315	0.266	0.458	5	−0.343 [†]
55555*	−0.659	0.365	555555	−0.728	0.337	5	−0.069 [†]

*For EQ-4D states, self-care dimension (the second digital of standard EQ-5D health state) was removed.

[†]Significant at 0.05 level.

changing from level 2 to level 3 in vision (β : −0.013), which was not statistically significant. Notably, many coefficients were not significant at the 0.05 level, especially for level-2 and level-3 variables. Using the coefficients estimated in models 4D and 5D_ARM1, the estimated coefficient for λ was 0.902 and 0.841, respectively. This suggests that bolting on the self-care and vision dimension led to a shrinkage of the main effects by around 10% and 15%, respectively. In both arms, the scaling factor models outperformed the standard models in all statistics, for example, the MAE of the standard models (arm 1: 0.128, arm 2: 0.134) were more than twice the size of the MAE of the scaling factor model (arm 1: 0.055, arm 2: 0.058).

Similarly, scaling factor models (see Table 4) also performed better using CALE model specifications. Compared with the additive models, the CALE models performed better both in terms of model prediction and model fit, for both arms. The effect was more obvious in arm 1, ie, when using EQ-4D data to predict

EQ-5D data. In arm 2, the difference was marginal (eg, MAE of additive model: 0.058, MAE of CALE model: 0.057). Based on the results of the CALE models, the bolt-on item had very different importance in these 2 arms, ie, self-care had the lowest dimension weight, and vision had the largest dimension weight.

Discussion

Consistent with previous studies,¹⁶ our study found that the effect of bolt-on items to EQ-5D on the valuation of resultant health states varies with the level of problems of the bolt-on. The effect was consistent across 2 arms with different bolt-on items and a different number of core dimensions. Specifically, no or mild problems increased health-state value and extreme problems decreased health-state values. For example, with EQ-4D as the core, adding a level-1 self-care problems as the bolt-on increased

Table 3. Additive model results.

	Arm 1			Arm 2		
	Model_4D	Model_5D_ARM1	Model_5D_SF	Model_5D_ARM2	Model_6D	Model_6D_SF
Model coefficients						
INTERCEPT	0.054*	0.023*	0.022*	0.055	0.014*	0.064
λ			0.902			0.841
MO2	0.073	0.037*	0.073 [†]	0.013*	0.048	0.013 [†]
MO3	0.058	0.065	0.058 [†]	0.078	0.021*	0.078 [†]
MO4	0.152	0.135	0.152 [†]	0.132	0.130	0.132 [†]
MO5	0.073	0.066	0.073 [†]	0.066	0.071	0.066 [†]
SC2		0.019*	0.025*	0.026*	0.054	0.026 [†]
SC3		0.044	0.043*	0.025*	0.020*	0.025 [†]
SC4		0.187	0.195	0.143	0.115	0.143 [†]
SC5		0.025*	0.049*	0.061	0.022*	0.061 [†]
UA2	0.039*	0.053	0.039 [†]	0.009*	0.005*	0.009 [†]
UA3	0.060	0.025*	0.060 [†]	0.041	0.041*	0.041 [†]
UA4	0.195	0.173	0.195 [†]	0.147	0.127	0.147 [†]
UA5	0.089	0.116	0.089 [†]	0.122	0.062	0.122 [†]
PD2	0.056	0.045	0.056 [†]	0.039	0.063	0.039 [†]
PD3	0.042*	0.039*	0.042 [†]	0.061	0.040*	0.061 [†]
PD4	0.203	0.201	0.203 [†]	0.216	0.178	0.216 [†]
PD5	0.109	0.063	0.109 [†]	0.073	0.037*	0.073 [†]
AD2	0.039*	0.059	0.039 [†]	0.030*	0.058	0.030 [†]
AD3	0.077	0.025*	0.077 [†]	0.111	0.084	0.111 [†]
AD4	0.235	0.224	0.235 [†]	0.165	0.168	0.165 [†]
AD5	0.121	0.143	0.121 [†]	0.117	0.065	0.117 [†]
VI2					0.027*	0.063
VI3					−0.011*	−0.013*
VI4					0.142	0.168
VI5					0.253	0.277
Model performances						
Log-likelihood		−2174.2	−1920.1		−1449.5	−1478.5
Pearson R		0.918	0.985		0.922	0.981
CCC		0.918	0.984		0.922	0.981
MAE		0.128	0.055		0.134	0.058
RMSE		0.153	0.068		0.170	0.082

Note. Each dimension has four dummy variables, eg, MO3 is the dummy variable for having moderate problems in mobility dimension. CCC, indicates concordance correlation coefficient; MAE, mean absolute errors; RMSE, root mean squared errors; SF, scaling factor; λ , scaling factor coefficient.

*Not significant at 0.05 level.

[†]Estimates taken from Model_4D and Model_5D_ARM2 respectively.

the health-state value; with EQ-5D as the core, adding a level-5 vision problems as the bolt-on resulted in a decrease in health-state values. More importantly, adding a bolt-on resulted in decreased coefficient values for core dimensions. This trend suggests that disutility associated with problems decreases or shrinks when the number of health dimensions increases. Hence, a scaling factor parameter for adjusting the effects of the dimensions in the core descriptive system sounded necessary and was proven to perform well in this study.

Importantly, our study discovered that empirically estimated dimensional disutility weights for predicting the value sets of a descriptive system can give accurate estimates for the values of

bolt-on health states when a scaling factor is used to adjust those weights. The better performance of the scaling factor model was validated in both EQ-5D-5L and the modified EQ-4D health states, with 2 different bolt-on items, respectively. This means that bolt-on value sets could be estimated based on standard EQ-5D value sets. For example, we could conduct a small valuation study of selected bolt-on health states and then use the data and the published dimensional weights of a standard EQ-5D value set to estimate the bolt-on value set. Such a value set is not just resource saving; it would preserve all the dominance relationships among the EQ-5D health states. The use of scaling factor model is not without caveat. So far, we hypothesized that (1) the scaling

Table 4. CALE model results.

	Arm 1			Arm 2		
	Model_4D	Model_5D_ARM1	Model_5D_SF	Model_5D_ARM2	Model_6D	Model_6D_SF
Model coefficients						
INTERCEPT	0.066	0.018*	0.027*	0.053	0.014*	0.049
λ			0.902			0.845
VI					0.388	0.441
MO	0.341	0.299	0.341 [†]	0.293	0.266	0.293 [†]
SC		0.299	0.330	0.257	0.213	0.257 [†]
UA	0.386	0.358	0.386 [†]	0.315	0.253	0.315 [†]
PD	0.404	0.358	0.404 [†]	0.397	0.330	0.397 [†]
AD	0.476	0.444	0.476 [†]	0.415	0.379	0.415 [†]
L2	0.113	0.135	0.113 [†]	0.074	0.145	0.074 [†]
L3	0.262	0.241	0.262 [†]	0.263	0.259	0.263 [†]
L4	0.757	0.748	0.757 [†]	0.737	0.705	0.737 [†]
Model performances						
Log-likelihood		−1461.5	−1464.7		−1489.5	−1498.2
Pearson R		0.979	0.988		0.968	0.983
CCC		0.979	0.988		0.966	0.982
MAE		0.058	0.045		0.080	0.057
RMSE		0.076	0.059		0.106	0.074

AD indicates anxiety/depression; CALE, cross-attribute level effects; CCC, concordance correlation coefficient; L2, level 2; L3, level 3; L4, level 4; MAE, mean absolute errors; MO, mobility; PD, pain/discomfort; RMSE, root mean squared errors; SF, scaling factor; λ , scaling factor coefficient; SC, self-care; UA, usual activities; VI, vision. *Not significant at 0.05 level.

[†]Estimates taken from Model_4D and Model_5D_ARM2 respectively.

effect is universal to all existing dimensions of the core and (2) there are no significant interactions between bolt-on and EQ-5D dimensions. Based on the first hypothesis, we added only 1 scaling parameter in the models, which seems to work well in the tested cases. However, multiple, dimension-specific scaling parameters may better predict bolt-on health states given that the information content reduction caused by the bolt-on may not be identical to all core dimensions. Yet, allowing multiple scaling parameters could easily alter the dominance relationships between health states. Based on the hypothesis that effects of EQ-5D dimensions do not interact with each other in modeling of valuation data, we used a main-effects model similar to those used to predict EQ-5D value sets. However, it is possible that bolt-on and some of the EQ-5D dimensions interact and the interactions should be specified in the model. Therefore, future studies should relax those assumptions and investigate alternative model specifications. Despite the consistent better performance of the scaling factor model, we found that the prediction accuracy was higher for arm 1. This may be due to a smaller number of dimensions or different bolt-on items.

The main reasons for the good performance of the scaling factor model could be, on one hand, by taking the coefficients of another model, only 3 (for CALE model specification) or 6 (for additive model specification) parameters need to be estimated for a bolt-on to the EQ-5D-5L. In such a case, the modeling process is well powered by a sample size of 300; in contrast, 10 and 24 parameters need to be estimated using the default models. On the other hand, by design, the scaling factor model utilizes more data (ie, the coefficients of the scaling factor models were derived from two student samples while the coefficients of the default models

were estimated from only one student sample). It should be acknowledged that when using the scaling factor model to estimate bolt-on value sets in reality, investigators need to collect bolt-on valuation data from a new population sample that may be drawn using a sampling method and studied using a valuation protocol different from the original EQ-5D valuation study. In this case, although the true relative effects of the core 5 dimensions and bolt-on on the values of health states may be constant, the scaling factor model may not perform well because sample characteristics and valuation methods may affect health-state values. It is unfortunate that our study did not allow us to assess the robustness of the scaling factor model to variability in study samples and valuation design. Further study should evaluate the performance of the scaling factor model using EQ-5D value sets that are adopted in the practice of health technology assessment.

CALE models have been found to be superior in terms of out-of-sample prediction accuracy for both original and modified EQ-5D cores compared with the additive model,⁷ although the generalizability of these findings needs to be investigated. Therefore, it is not surprising that the CALE model specification produced better results for the scaling factor model than the additive model. The CALE models offer an attractive opportunity to investigate the impact of adding a bolt-on dimension to an existing health descriptive system because they contain fewer parameters and therefore need smaller sample sizes in valuation studies. Nevertheless, the additive model could be more useful when the scaling factor model is used to estimate bolt-on value sets because most of the existing EQ-5D value sets are produced using it. The design of the scaling factor model requires its

specification for the effects of individual dimensions to follow that of the model used to produce the standard EQ-5D value sets, so that the parameters in the latter can be directly plugged into the former. If the CALE model specification is chosen for the scaling factor model for producing bolt-on value sets for an EQ-5D value set that is produced using the additive model, one would have to reestimate the EQ-5D value set using the CALE model first, which is not only more work but may also create the issue of dominance inconsistency between original and bolt-on value sets.

There are several limitations that need to be recognized. First, we asked each respondent to value 2 types of health states (ie, EQ-4D and EQ-5D or EQ-5D and EQ-6D) that shared the same core dimensions. Although this design allowed us to study the effect of the bolt-on directly, it is possible that values of the second type of health states were biased. Second, we collected data from university students instead of the general public. The homogeneity of students' backgrounds allowed us to easily achieve 2 comparable samples, and it is easier for young and well-educated persons to complete a large number of valuation tasks. However, the findings may not be generalized to the general populations. Third, we only studied the effect of self-care on EQ-4D and the vision bolt-on on EQ-5D-5L. Our findings may not be generalized to other dimensions or bolt-on candidates. Therefore, future studies should test other well-established bolt-on items such as cognition and hearing in general population samples. Last but not least, the use of random intercepts may be considered an overly simplistic approach to addressing respondent-level heterogeneity. However, more complex random effects, such as separate random slopes in dimension severity, cannot currently be modeled within the analysis framework used in this study.

Conclusions

The scaling factor model offers a methodologically viable and low-cost option for producing bolt-on value sets. Further empirical testing using other valuation methods, such as discrete choice experiment, other bolt-on items, other model specifications, and more relevant study populations is needed to further evaluate the potential of this novel approach.

Author Disclosures

Author disclosure forms can be accessed below in the [Supplemental Material](#) section.

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